3.5.1 Newton's laws of motion

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	Newton's three laws of motion	
(b)	linear momentum; $p = mv$; vector nature of momentum	
(c)	net force = rate of change of momentum*;* $F = \dfrac{\Delta p}{\Delta t}$	Learners are expected to know that $F = ma$ is a special case of this equation. *M2.1, M3.9*
(d)	impulse of a force; impulse = $F\Delta t$	
(e)	impulse is equal to the area under a force–time graph.	Learners will also be expected to estimate the area under non-linear graphs. HSW3 Using a spreadsheet to determine impulse from F–t graph. *M3.8, M4.3*